SIGHTS, SOUNDS & SCENTS

Multisensory enrichment of AI-generated advertisements enhances engagement and modulates visual attention

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Background

Human experience is inherently multisensory, yet advertising research has lagged behind. Short-form social media video is now a dominant format with largely untapped multisensory potential. We examined whether semantically congruent sound and scent enhance viewer perception and engagement.

Research objective

1. Does the number and nature of additional stimuli (audio, olfaction) influence the perception of the visual advertisement stimuli?
2. Are differences in advertisement perception and the effect of additional sensory modalities observable with eye tracking analysis?

Methods

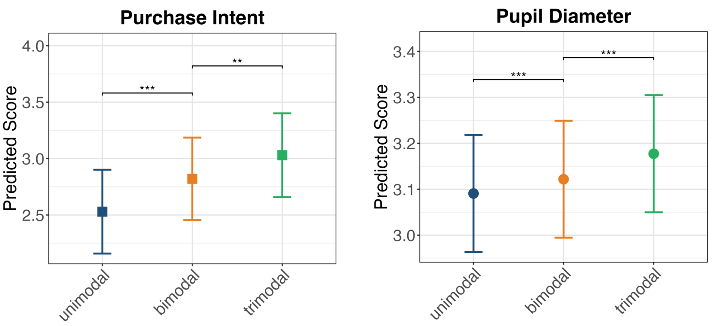
Five AI-generated **advertisement** videos (SORA) each covering a different topic, were embedded in a vertical social-media-like feed and shown to 45 participants with added olfactory and/or auditory enrichment, treated as number of modalities for analysis:

Unimodal
Bimodal + /
Trimodal + +

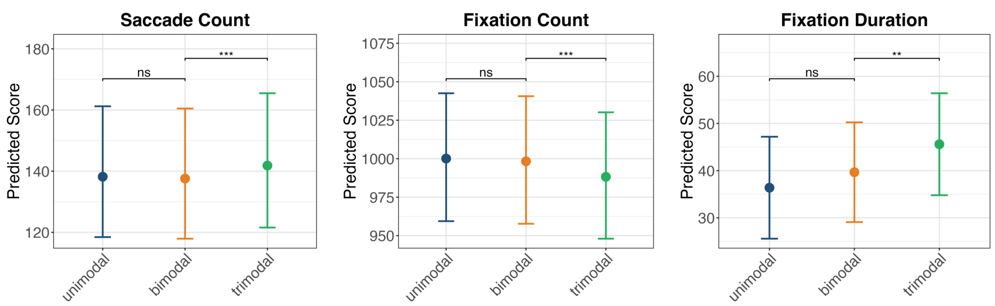
Participants rated **video quality**, **product liking**, and **purchase intent,** while **gaze** was recorded with wearable eye-tracking glasses. Linear mixed models estimated modality effects on behavioral and eye-tracking outcomes, with age and gender as covariates and participant as a random intercept.

Results

Purchase intent and pupil dilation both increase stepwise from unimodal to trimodal



Trimodal viewing raised saccade count and fixation duration but reduced fixation count



Take-away

☛ Multisensory enrichment enhances both subjective engagement and physiological arousal

☛ More active exploration paired with more selective attentional allocation, suggesting deeper cognitive processing of advertising content

☛ Potential of congruent multisensory cues to enhance advertising effectiveness in digital environments

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